DERIVED HOM AGAINST A COMPACT BIMODULE WITH REPRESENTABLE DUAL IS DERIVED TENSOR WITH THE DUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the (B, A)-bimodule structure on S from the dg-category composition, the comparison map c_N and its functoriality, the derived transformation through property-(P) and property-(I) replacements, and the two-step compactness reduction to N = B[k] closed by assumption (2) are all retained verbatim.

Editorial note. The label lemma-RHom-is-tensor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The differential graded category structure and bimodule actions. Same chapter, Example example-dgm-dg-cat and the module structures of section-restriction. Supplies the composition on Hom complexes that gives S = Hom(T, B) its (B, A)-bimodule structure and the counit-type multiplication defining the comparison map c_N.
- Compactness and the two-term filtrations of property (P). Same chapter, Lemma lemma-property-P-sequence; the compact-object formalism of proposition-compact. Lets both sides, being exact functors of triangulated categories with T compact, be checked on finite filtrations and finally on shifts of B.

Source reference. Original reference 09LC: sources/stacks-dga/pinned-original-source.tex, source lines 3525–3525. Prerequisite pointer invoked at proof line(s) 5806; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09KL: sources/stacks-dga/pinned-original-source.tex, source lines 2425–2425. Prerequisite pointer invoked at proof line(s) 5827; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let T be a differential graded (A, B)-bimodule. Assume*

(1) *T defines a compact object of $D(B, d)$, and*
(2) *$S = \mathrm{Hom}_{Mod^{dg}_{(B,d)}}(T, B)$ represents $R\mathrm{Hom}(T, B)$ in $D(A, d)$.*

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Then S has a structure of a differential graded (B, A)-bimodule and there is an isomorphism

$$N \otimes^{\mathbf{L}}_B S \longrightarrow R\operatorname{Hom}(T, N)$$

functorial in N in $D(B, d)$.

Proof. Write $\mathcal{B} = \operatorname{Mod}^{dg}_{(B,\mathrm{d})}$. The right A-module structure on S comes from the map $A \to \operatorname{Hom}_{\mathcal{B}}(T, T)$ and the composition $\operatorname{Hom}_{\mathcal{B}}(T, B) \otimes \operatorname{Hom}_{\mathcal{B}}(T, T) \to \operatorname{Hom}_{\mathcal{B}}(T, B)$ defined in Example 09LC. Using this multiplication a second time there is a map

$$c_N : N \otimes_B S = \operatorname{Hom}_{\mathcal{B}}(B, N) \otimes_B \operatorname{Hom}_{\mathcal{B}}(T, B) \longrightarrow \operatorname{Hom}_{\mathcal{B}}(T, N)$$

functorial in N. Given N we can choose quasi-isomorphisms $P \to N \to I$ where P, resp. I is a differential graded B-module with property (P), resp. (I). Then using c_N we obtain a map $P \otimes_B S \to \operatorname{Hom}_{\mathcal{B}}(T, I)$ between the objects representing $S \otimes^{\mathbf{L}}_B N$ and $R\operatorname{Hom}(T, N)$. Clearly this defines a transformation of functors c as in the lemma.

To prove that c is an isomorphism of functors, we may assume N is a differential graded B-module which has property (P). Since T defines a compact object in $D(B, \mathrm{d})$ and since both sides of the arrow define exact functors of triangulated categories, we reduce using Lemma 09KL to the case where N has a finite filtration whose graded pieces are direct sums of $B[k]$. Using again that both sides of the arrow are exact functors of triangulated categories and compactness of T we reduce to the case $N = B[k]$. Assumption (2) is exactly the assumption that c is an isomorphism in this case. □